



Original Articles

Machine perfusion prevents early tumor recurrence in liver transplantation for hepatocellular carcinoma: a multicenter retrospective cohort study

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ABSTRACT

There is controversy regarding whether the machine perfusion (MP) prevents hepatocellular carcinoma (HCC) recurrence after liver transplantation (LT). This study aimed to explore the impact of MP on HCC recurrence. We included data from the United Network for Organ Sharing (UNOS) database on HCC patients spanning from April 2015 and January 2024. Patients who received MP-preserved donor were propensity score-matched (PSM) 1:1 with those who received static cold storage (SCS)-preserved donor. The primary outcome was recurrence-free survival (RFS) rate. Cox regression models were used to identify predictors of RFS. Subgroup analyses assessed the role of MP across various groups and to identify the beneficial group. The PSM cohort comprised 411 HCC recipients with MP-preserved donors and 411 with SCS-preserved donors. The 2-year RFS rate was significantly higher in the MP group compared to the SCS group (96.5 % vs. 89.7 %, $P = 0.008$). Univariate and multivariate Cox regression analyses revealed that MP preservation method was an independent predictor of RFS [adjusted hazard ratio (aHR), 0.158; 95 % confidence interval (CI), 0.048–0.523; $P = 0.003$]. In the beneficial group, defined by the cumulative of MP-enhancing factors, MP improved RFS comparable to SCS group (96.7 % vs. 76.2 %, $P = 0.001$). MP effectively prevents HCC recurrence post-LT. MP should be used more proactively in the beneficial groups, including those with hemodynamically unstable donors and high tumor burden, to improve the survival outcomes.

Abbreviations: HCC, hepatocellular carcinoma; LT, liver transplantation; IRI, ischemia reperfusion injury; WIT, warm ischemia time; CIT, cold ischemia time; MP, machine perfusion; NMP, normothermic machine perfusion; HMP, hypothermic machine perfusion; UNOS, United Network for Organ Sharing; RFS, recurrence-free survival; STAR, standard transplant analysis and research; DCD, donation after circulatory death; BMI, body mass index; TBil, total bilirubin; CPR, cardiopulmonary resuscitation; MELD, model for end-stage liver disease; AFP, alpha-fetoprotein; MI, multiple imputation; PSM, propensity score matching; IQR, interquartile ranges; OS, overall survival; HR, hazard ratios; CI, confidence intervals; ROC, receiver operating characteristic; IFLT, ischemia-free liver transplantation; ROS, reactive oxygen species; VEGF, vascular endothelial growth factor; MMP-9, matrix metalloproteinase-9; TIT, total ischemia time; RCT, randomized controlled trials.

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1. Introduction

Hepatocellular carcinoma (HCC) is a highly malignant cancer and ranks as the fourth leading cause of cancer related deaths worldwide [1, 2]. Among the available options, liver transplantation (LT) has emerged as the one of the most effective therapeutic approaches for HCC, offering the potential for curative outcomes [3]. However, postoperative tumor recurrence is a major challenge and concern for LT in patients with HCC [4]. Extensive studies have identified several risk factors, including tumor burden and biology, which are associated with HCC recurrence [5]. In addition to these factors, increasing studies and clinical evidence unmask that ischemia reperfusion injury (IRI) promotes the recurrence of HCC after LT [6]. Experimental models have demonstrated that IRI induced changes in the hepatic microenvironment promote HCC recurrence [7]. Meanwhile, clinical studies have also shown that extended durations of warm ischemia time (WIT) and cold ischemia time (CIT) are correlated with HCC recurrence [8,9].

The machine perfusion (MP) techniques, including normothermic machine perfusion (NMP) and hypothermic machine perfusion (HMP), are highly regarded for their ability to mitigate IRI and enhance graft function [10]. Current studies on the effectiveness of MP technology in reducing HCC recurrence by mitigating IRI remains a topic of ongoing debate [11–14]. The reliance on single center experiences and the lack of precise analysis of potential beneficiary subgroup highlight the necessity for further in-depth research.

In the present study, we initially analyzed the United Network for Organ Sharing (UNOS) database and evaluated the recurrence-free survival (RFS) rate of HCC patients treated with MP. Subsequently, we conducted a subgroup analysis to further assess the specific effects of MP on beneficial subgroup. This study provides valuable insights into the clinical significance of MP and its potential impact on specific subgroups, paving the way for improved outcomes in HCC recipients.

2. Patients and methods

2.1. Study design and patient population

This study utilized the July 2024 release of the standard transplant analysis and research (STAR) files from the UNOS database and included adult recipients diagnosed with HCC treated with LT between April 2015 and January 2024 ($n = 14,335$). The included and exclusion criteria is shown in Fig. 1.

The study variables included donor, recipient, and tumor parameters. The donor variables were age, gender, body mass index (BMI), serum sodium, donation after circulatory death (DCD) donors, donor cardiopulmonary resuscitation (CPR) and CIT. The recipient variables included age, gender, BMI, preoperative laboratory model for end-stage liver disease (MELD) score. Tumor parameters were pretransplant alpha-fetoprotein (AFP), size of HCCs, number of lesions, satellite lesion, tumor differentiation, and vascular invasion.

Details regarding ethics statement, HCC recurrence, method for handling missing values, propensity score matching, definition of the beneficial subgroup, sensitivity analyses, and statistical analysis are provided in the supplementary materials.

3. Results

3.1. Patient characteristics

The comparison of clinicopathological features is summarized in Table 1. Briefly, 10,121 male and 3,043 female recipients were included. Donors in the MP group were significantly older (median age 49.0 vs. 45.0 years) and included a higher proportion of male (68.4 % vs. 60.7 %), DCD donors (51.2 % vs. 11.1 %) and longer CIT (median values 10.6 vs. 5.8 h). The SCS group exhibited a significantly longer follow-up duration compared to the MP group (median values 12.1 vs. 59.4

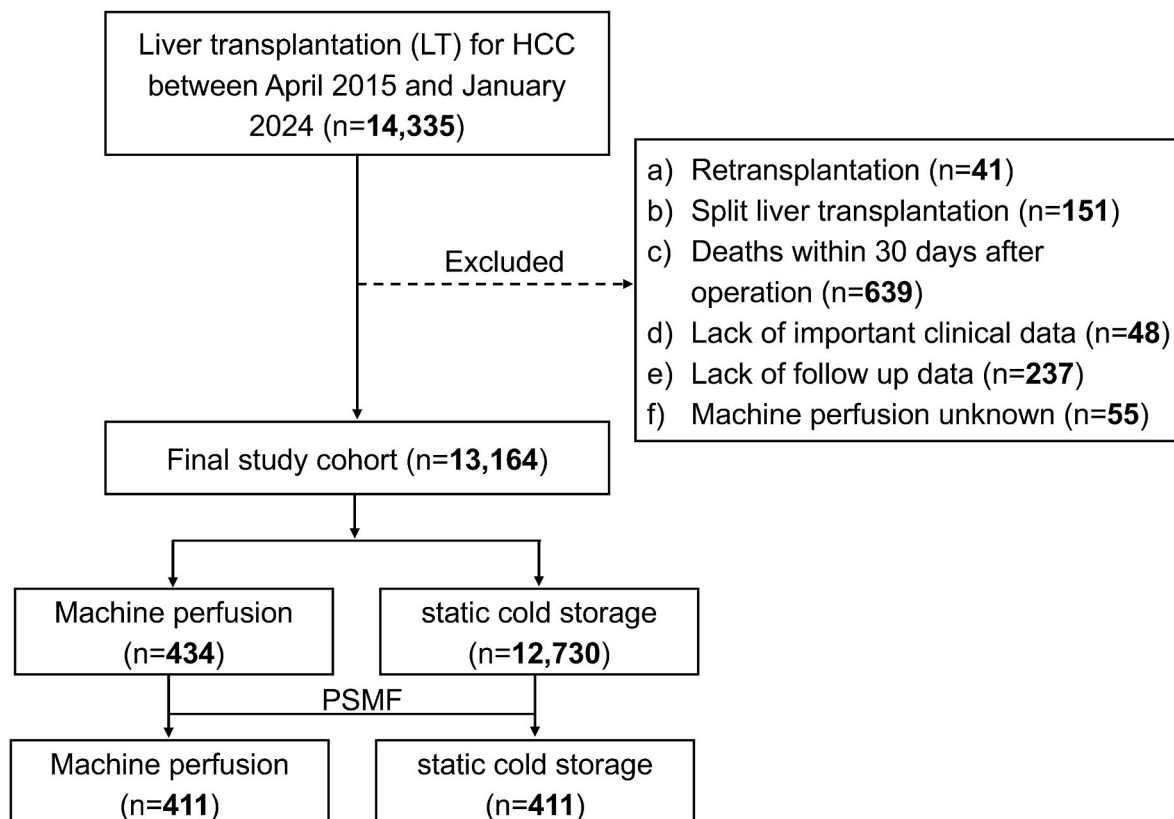


Fig. 1. Flowchart of patient selection procedures.

Table 1

Patient characteristics at baseline of the study cohort before and after propensity score matching.

	Entire cohort			Propensity-matched cohort			
	Machine perfusion (n = 434)	Static cold storage (n = 12,730)	P-value	Machine perfusion (n = 411)	Static cold storage (n = 411)	P-value	SMD*
Donor characteristics							
Donor age (years)	49.0 (36.0–59.0)	45.0 (31.0–57.0)	<0.001	48.0 (35.0–58.0)	46.0 (34.0–57.0)	0.385	0.059
Gender (male)	297 (68.4 %)	7724 (60.7 %)	0.001	278 (67.6 %)	274 (66.7 %)	0.824	0.021
Donor BMI (kg/m ²)	27.6 (24.0–33.0)	27.5 (23.9–32.1)	0.199	27.7 (24.0–33.1)	27.0 (23.6–31.8)	0.135	0.168
Donor serum creatinine (mg/dL)	1.0 (0.7–2.0)	1.1 (0.8–2.0)	0.009	1.0 (0.7–2.0)	1.0 (0.7–1.8)	0.475	0.097
Donor total bilirubin (mg/dL)	0.6 (0.4–0.9)	0.6 (0.4–1.0)	0.008	0.6 (0.4–0.9)	0.5 (0.4–0.8)	0.262	0.010
Donor serum sodium (mmol/L)	147.5 (141.0–152.0)	148.0 (143.0–153.0)	0.016	147.0 (141.0–152.0)	147.0 (142.0–152.0)	0.690	0.035
DCD donors	222 (51.2 %)	1418 (11.1 %)	<0.001	199 (48.4 %)	205 (49.9 %)	0.727	0.029
Donor CPR	234 (54.9 %)	6589 (53.4 %)	0.531	223 (55.2 %)	227 (57.3 %)	0.545	0.045
Donor antihypertensives	144 (33.2 %)	3931 (30.9 %)	0.311	191 (50.8 %)	185 (45.0 %)	0.674	0.057
Cold ischemia time (hours)	10.6 (7.7–15.0)	5.8 (4.6–7.1)	<0.001	10.6 (7.7–14.7)	5.5 (4.5–7.3)	<0.001	0.990
Recipient characteristics							
Age at transplant (years)	64.0 (59.0–68.0)	63.0 (58.0–67.0)	<0.001	64.0 (59.0–68.0)	65.0 (60.0–68.0)	0.139	0.112
Gender (male)	331 (76.3%)	9790 (76.9 %)	0.757	313 (76.2 %)	322 (78.3 %)	0.506	0.052
Recipient BMI (kg/m ²)	29.2 (25.5–32.3)	28.9 (25.5–32.7)	0.832	29.2 (25.7–32.4)	28.7 (25.5–32.3)	0.375	0.048
Recipient total bilirubin (mg/dL)	0.6 (0.4–0.9)	0.8 (0.5–3.0)	<0.001	0.6 (0.4–0.8)	0.6 (0.6–1.0)	0.078	0.060
Preoperative MELD score	11.0 (8.0–16.0)	11.0 (8.0–17.0)	0.273	10.0 (8.0–14.0)	10.0 (7.0–13.0)	0.125	0.114
Waiting time (days)	247.0 (184.5–451.3)	262.0 (192.0–422.0)	0.315	257.0 (188.5–454.5)	242.0 (185.0–351.0)	0.060	0.224
Tumor parameter							
Pretransplant AFP (ng/mL)	6.0 (3.0–11.3)	6.0 (4.0–15.0)	0.112	6.0 (3.0–12.0)	7.0 (4.0–15.5)	0.135	0.005
Size biggest HCC lesion (cm)	1.8 (0.0–2.4)	2.0 (0.0–2.6)	0.101	1.9 (0.0–2.4)	1.2 (0.0–2.4)	0.233	0.112
Overall size all HCCs (cm)	2.0 (0.0–2.4)	2.1 (0.0–3.1)	0.271	2.0 (0.0–3.0)	1.3 (0.0–2.8)	0.328	0.056
Number of lesions (single)	378 (87.1 %)	10772 (84.6 %)	0.159	359 (87.3 %)	353 (85.9 %)	0.490	0.048
Meeting the Milan criteria	359 (82.7 %)	10692 (84.0 %)	0.478	345 (83.9 %)	340 (82.7 %)	0.640	0.033
Satellite lesions	21 (4.8 %)	661 (5.2 %)	0.744	19 (4.6 %)	18 (4.4 %)	1.000	0.012
Tumor differentiation			0.319			0.237	0.093
Complete tumor necrosis	102 (23.5 %)	3230 (25.4 %)		101 (24.6 %)	118 (28.7 %)		
Well	77 (17.7 %)	2584 (20.3 %)		74 (18.0 %)	69 (16.8 %)		
Moderate	224 (51.6 %)	6033 (47.4 %)		208 (50.6 %)	199 (48.4 %)		
Poor	31 (7.1 %)	883 (6.9 %)		28 (6.8 %)	25 (6.1 %)		
Vascular invasion			0.373			0.554	0.038
None	363 (83.6 %)	10949 (86.0 %)		349 (84.9 %)	354 (86.1 %)		
Microvascular invasion	62 (14.3 %)	1545 (12.1 %)		54 (13.1 %)	51 (12.4 %)		
Macrovascular	9 (2.1 %)	236 (1.9 %)		8 (1.9 %)	6 (1.5 %)		
Duration of follow-up (months)	12.1 (12.0–12.2)	59.4 (58.4–60.4)	<0.001	12.2 (11.9–12.4)	12.9 (11.9–13.9)	0.075	0.069

Note: Results are presented as n (%) or median (IQR). Values in bold represent statistically significant differences.

Abbreviations: BMI, body mass index; DCD, donation after circulatory death; CPR, cardiopulmonary resuscitation; MELD, model for end-stage liver disease; AFP, alpha-fetoprotein; HCC, hepatocellular carcinoma; SMD, standardized mean difference.

*SMD values are presented as absolute values.

months), primarily reflects the recent adoption timeline of MP technology.

The balance between the two groups improved significantly following the application of PSM (Table 1). The variable CIT was not matched due to the characteristics of MP technique, resulting in significantly longer CIT in the MP group compared to the SCS group (median values 10.6 vs. 5.5 h). Other baseline data were well balanced in the

matched cohort and shown in Table 1.

3.2. Relationship between machine perfusion and post-transplant outcomes

In the overall cohort, no significant differences were observed in 2-year RFS and OS between the MP and SCS groups ($P = 0.085$, $P =$

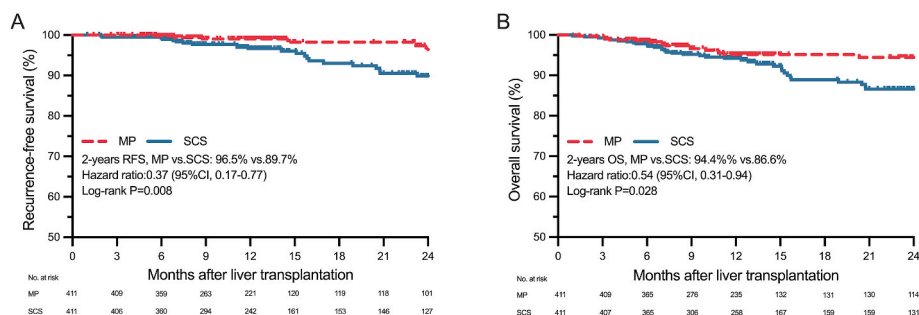


Fig. 2. Recurrence and overall survival curves for MP vs. SCS preserve after the propensity score matched. (A) (B). RFS and OS curves in recipients MP vs. SCS using log-rank tests.

0.314; Fig. S2A and B). After matching, the 2-year RFS rate in the MP group was 96.5 %, which was significantly higher than that in the SCS group at 89.7 % ($P = 0.008$; Fig. 2A). The 2-year OS rate in the MP group was also higher than the SCS group ($P = 0.028$; Fig. 2B). No significant differences in 2-year RFS or OS were observed between subgroups stratified by perfusion type, including normothermic and hypothermic ($P = 0.579$, $P = 0.839$; Fig. S3A and B).

To identify potential risk factors for RFS, univariable and multivariable Cox regression analyses were conducted. According to multivariable Cox regression analysis, MP was protective factor for RFS in comparison with SCS (adjusted HR, 0.158; 95 % CI, 0.048–0.523; $P = 0.003$; Table S3, Table 2). Similarly, MP was also independent risk factors for OS (adjusted HR, 0.316; 95 % CI, 0.160–0.627; $P < 0.001$; Table S3, Table S4).

3.3. Machine perfusion significantly improves post-transplant outcomes in beneficial subgroup

To identify MP efficacy factors, we conducted subgroup analyses. ROC curve analysis determined optimal cutoff values of 13 ng/mL for AFP and 28 kg/m² for BMI (Fig. S4A–B). Subgroup analyses demonstrated significantly improved outcomes with MP in patients with AFP ≥ 13 ng/mL, BMI < 28 kg/m², CIT ≥ 10 h, and beyond Milan criteria (Fig. S5). Based on the identified MP-enhancing factors, patients were stratified into beneficial (2–4 points) and non-beneficial (0–1 points) subgroups (Table S5). Baseline characteristics are summarized in Table S6–7. The non-beneficial group demonstrated a superior 2-year RFS compared the beneficial group ($P < 0.001$; Fig. S6A). In the non-beneficial subgroup, MP did not significantly improve patient outcomes compared to the SCS group ($P = 0.286$; Fig. 3A). In contrast, in the beneficial subgroup, the 2-year RFS in the MP group was 96.7 %, which showed a significant improvement compared to 76.2 % in the SCS group ($P = 0.001$; Fig. 3B). Further stratification of the beneficial group revealed that MP treatment significantly improved 2-year RFS, achieving levels comparable to the non-beneficial group ($P = 0.317$; Fig. S6B).

In terms of OS, the non-beneficial group demonstrated a superior OS compared to the beneficial group ($P = 0.015$; Fig. S6C). In the non-beneficial subgroup, MP did not significantly improve patient outcomes ($P = 0.250$; Fig. 3C). In the beneficial subgroup, the 2-year OS in the MP group was 93.7 %, which showed a significant improvement compared to 76.3 % in the SCS group ($P = 0.022$; Fig. 3D). Further stratification revealed that MP treatment significantly improved 2-year OS achieving levels of the non-beneficial group ($P = 0.310$; Fig. S6D).

3.4. Sensitivity analyses

Baseline characteristics for all sensitivity analyses are summarized in Table S8–10. After excluding participants enrolled before 2022, the result was consistent with the primary outcome, demonstrating a

significantly lower 2-year RFS for the SCS group at 82.6 % compared to 96.8 % in the MP group ($P = 0.003$; Fig. 4A). After adjusting for confounding factors, MP was identified as a protective factor against post-operative recurrence (adjusted HR, 0.121; 95 % CI, 0.015–0.962; $P = 0.046$, Table S11). In analyses accounting for follow-up duration, the result was consistent with the primary outcome and showed that the 2-year RFS of the MP group was 98.2 %, higher than that of the SCS group at 90.3 % ($P = 0.017$; Fig. 4B). Cox multivariate regression analysis further confirmed that MP was a protective factor (adjusted HR, 0.231; 95 % CI, 0.077–0.689; $P = 0.009$; Table S11). After excluding cases with missing data, the results showed that the 2-year RFS in the MP group was 97.6 %, significantly higher than that in the SCS group at 86.6 % ($P < 0.001$; Fig. 4C). Multivariate Cox regression analysis indicated that MP significantly reduced the risk of recurrence (adjusted HR, 0.162; 95 % CI, 0.048–0.552; $P = 0.004$; Table S11).

Using competing risk analysis, the results similarly revealed that the cumulative recurrence rate was significantly higher in the SCS compared to the MP group (9.3 % vs. 1.7 %; $P = 0.006$; Fig. 4D). Fine and Gray's sub-distribution hazards regression model confirmed that MP was a protective factor (adjusted HR, 0.172; 95 % CI, 0.047–0.625; $P = 0.008$; Table S11).

Subgroup analyses as shown in Fig. S7, the results showed that MP was a protective factor in these subgroups. The stratified Cox multivariate regression analysis results are summarized in Table S12.

4. Discussion

The present study represents the largest known cohort to investigate the impact of MP on post-transplant HCC recurrence. Our findings demonstrated that MP technology significantly enhance RFS follow-LT. In addition, the MP treatment significantly improves RFS in beneficial subgroup.

The finding by Patel et al. showing no significant difference in 1-year patient survival between MP and SCS groups (93.6 % vs. 93.2 %) is inconsistent with our results [14]. This divergence may be attributable to differences in study populations - their MP cohort included older donors and a higher proportion of DCD livers. These reasons may also extend to explain the discrepancy with Rigo et al.'s findings [13]. The favorable 2-year survival rates observed in our study are consistent with international data, likely attributable to optimized patient selection and the unique curative potential of total liver replacement compared to partial resection [14,15].

The present findings are consistent with Mueller et al. and Tang et al. [11,12]. Mueller et al. showed a significantly lower HCC recurrence rate of 5.7 % in DCD livers treated with HOPE, compared to 25.7 % in unperfused DBD livers [11]. Tang et al. further explored the impact of MP techniques by introducing the ischemia-free liver transplantation (IFLT) technique, which revealed that IFLT significantly enhance RFS rate compared to conventional LT (86.7 % vs. 46.3 %) [12].

Another major finding of this study is the pronounced protective

Table 2
Multivariable Cox regression analyses of prognostic factors for the recurrence-free survival.

	Model 1			Model 2			Model 3		
	HR	95 % CI	P	HR	95 % CI	P	HR	95 % CI	P
Preservation method (MP vs. SCS)	0.255	0.087–0.747	0.013	0.231	0.078–0.689	0.009	0.158	0.048–0.523	0.003
Recipient BMI (kg/m ²)	0.894	0.819–0.975	0.011	0.886	0.811–0.968	0.007	0.876	0.799–0.962	0.005
Meeting the Milan criteria	2.254	1.069–5.959	0.035	2.581	1.094–6.087	0.030	2.421	0.991–5.912	0.052
Pre-transplant AFP (ng/mL)	–	–	–	1.003	1.001–1.005	0.005	1.003	1.001–1.005	0.007
CIT (10 h)	–	–	–	–	–	–	3.238	1.155–9.084	0.026

Note: Values in bold represent statistically significant differences.

Model 1: Adjusted for Preservation method, Recipient BMI, Donor gender, Tumor differentiation, Meeting the Milan criteria, Vascular invasion.

Model 2: Adjusted for Model 1 + Pre-transplant AFP + Satellite lesions.

Model 3: Adjusted for Model 2 + CIT + DCD donor + Donor CPR.

Abbreviations: HCC, hepatocellular carcinoma; LT, liver transplantation; MP, machine perfusion; SCS, static cold storage; BMI, body mass index; DCD, donation after circulatory death; CPR, cardiopulmonary resuscitation; AFP, alpha-fetoprotein; CIT, cold ischemia time.

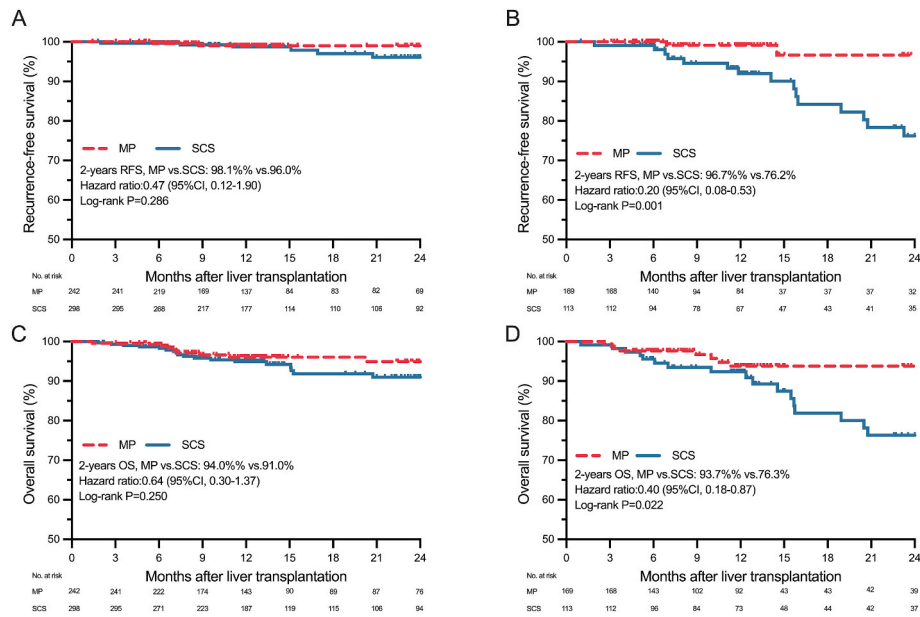


Fig. 3. Recurrence and overall survival curves for patients in non-beneficial and beneficial groups. (A). RFS curves in recipients in non-beneficial group; (B). RFS curves in recipients in beneficial group; (C). OS curves in recipients in non-beneficial group; (D). OS curves in recipients in beneficial group.

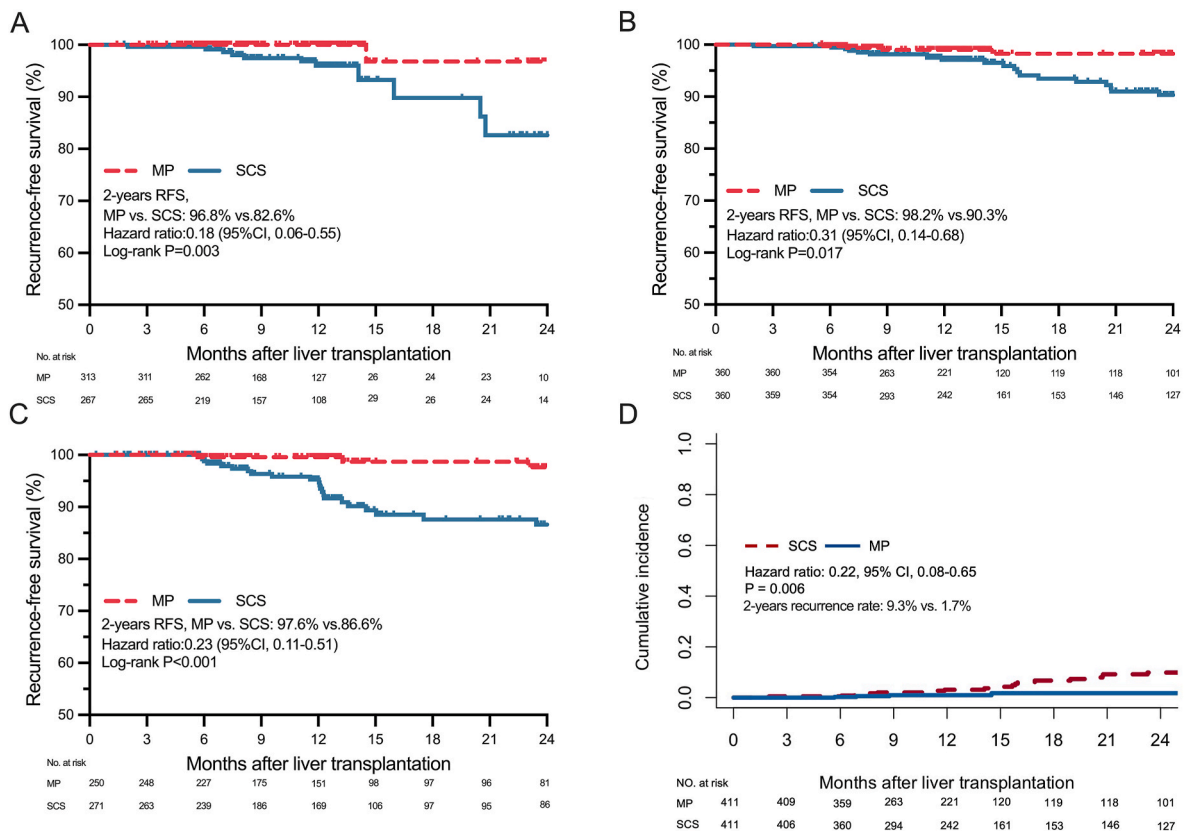


Fig. 4. Sensitivity analysis of recurrence survival rate. (A). RFS curves in recipients during 2022–2024; (B). RFS curves in recipients excluding HCC patients with a follow-up time of less than 6 months; (C). RFS curves in recipients excluding HCC patients with missing values in key variables; (D). Cumulative recurrence curves using the Fine and Gray model.

effect of MP in beneficial patients, as defined by the cumulative of MP-enhancing factors. These factors include tumor burden, as well as CIT ≥ 10 h. The superior efficacy of MP in patients with high tumor burden aligns with the findings of Philipp Dutkowski et al. [11]. Firstly, recipients with high tumor burden inherently have poorer post-transplant

outcomes, making them more likely to benefit from improved graft quality. Secondly, patients with high tumor burden are more prone to microvascular invasion. Microvascular invasion leads to increased circulating tumor cells (CTC) [16]. MP can improve graft quality by mitigating venous endothelial damage, which consequently reduces

post-transplant adhesion of CTC and effectively decreases tumor recurrence [17]. The superior efficacy of MP in recipients with CIT ≥ 10 h primarily stem from its ability to mitigate hepatic IRI. Building upon these findings, our study suggests that MP should be used more preferentially in beneficial subgroups.

The mechanisms by MP reduce HCC recurrence remain a topic of exploration and the widely accepted consensus is that MP could alleviate IRI. The primary driver of IRI is mitochondrial dysfunction, leading to reactive oxygen species (ROS) generation and oxidative stress [18–20]. Liver ischemia significantly accelerates tumor cell growth which associated with increased HIF-1 α expression [21]. Moreover, HIF-1 α stimulates angiogenesis by upregulating vascular endothelial growth factor (VEGF) and other pro-angiogenic factors, thus contributing to its aggressive progression [22,23]. Immune cells exacerbate liver injury and ROS production, creating a favorable microenvironment for cancer proliferation and recurrence [24]. Another important factor is the IRI induced upregulation of VEGF and matrix metalloproteinase-9 (MMP-9), which play complex roles in cancer progression by promoting angiogenesis and enhancing the invasive ability of tumor cells [25]. Additionally, chemokines induced by IRI may promote HCC progression through immune suppression by mobilizing regulatory T cells.

Evidence from a multicenter study demonstrated that a total ischemia time (TIT) exceeding 6 h were associated with a higher recurrence rate [26]. Furthermore, a large cohort study, which included over 9000 patients, revealed that prolonged WIT exhibited significantly higher recurrence rates [27]. Additionally, Yu et al. identified prolonged CIT exceeding 10 h as an independent risk factor for RFS [28]. These findings, supported by clinical evidence, indicate that IRI significantly contributes to tumor recurrence.

To address the challenges posed by IRI, various strategies have been explored, including pharmacological interventions, cell-based therapies, and ischemic preconditioning [24,29]. MP technology has become relatively mature and has proven effective in improving IRI [30]. Recent randomized controlled trials (RCT) have demonstrated its efficacy in reducing postoperative complications and improving graft function [31]. The benefits of MP include not only ameliorating IRI but also extending organ preservation time [32]. An additional advantage lies in its capacity to assess high-risk donor livers, enabling real-time evaluation of graft viability [33].

In our study, we did not carefully distinguish between different types of MP, including HMP and NMP. Additionally, the UNOS database lacked detailed records of MP usage, including its duration and sequence. Notably, our preliminary analysis found no significant difference between HMP and NMP. In previous studies, various types of MP have been demonstrated to alleviate IRI [34]. Several MP devices have been approved for clinical use in LT [25]. Therefore, in this study, we treated MP as a unified intervention to explore whether mitigating IRI could effectively reduce tumor recurrence. Future studies, prospective data collection with standardized MP protocols could help address these gaps and provide clearer guidance for clinical application.

This study has several limitations for improvement. Firstly, as this study is based on a retrospective design, it may be subject to inherent biases. Secondly, as MP is a relatively recent advancement, the current analysis is only sufficiently powered to evaluate short-term outcomes, the long-term benefits require further investigation with extended follow-up. Finally, as this study is based on the UNOS database and reflects characteristics of the U.S. population, the generalizability of the findings to other regions remains uncertain. Further studies with data from diverse populations are required to confirm the applicability of these findings.

In conclusion, our study confirmed MP effectively reduces the risk of HCC recurrence and identified that patients in the beneficial subgroups. Therefore, it is crucial to adopt a more proactive approach in applying MP to these patients. Our research may lay the groundwork for the precise selection of suitable candidates for MP, further optimizing its application and therapeutic potential.

CRediT authorship contribution statement

Wenzhi Shu: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Hao Chen:** Writing – original draft, Conceptualization. **Ruolin Wang:** Writing – original draft, Software. **Jiyong Song:** Methodology, Funding acquisition. **Rui Tang:** Writing – review & editing. **Guangdong Wu:** Writing – review & editing. **Lihan Yu:** Writing – review & editing, Visualization. **Xuan Tong:** Writing – review & editing. **Xiaojuan Wang:** Writing – review & editing, Methodology. **Yucheng Hou:** Data curation. **Wen long Zhao:** Methodology. **Lizhen Zhu:** Writing – review & editing. **Jun Yan:** Validation, Supervision, Resources, Project administration, Funding acquisition. **Qian Lu:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.canlet.2025.217970>.

References

- [1] Z.J. Brown, D.I. Tsilimigras, S.M. Ruff, et al., Management of hepatocellular carcinoma: a review, *JAMA Surg.* 158 (2023) 410–420.
- [2] L.Q. Cao, Y. Xie, J.S. Fleishman, et al., Hepatocellular carcinoma and lipid metabolism: novel targets and therapeutic strategies, *Cancer Lett.* 597 (2024) 217061.
- [3] X. He, S. Xu, L. Tang, et al., Insights into the history and tendency of liver transplantation for liver cancer: a bibliometric-based visual analysis, *Int. J. Surg.* 110 (2024) 406–418.
- [4] X. Zhang, C. Chen, Y. Wang, et al., Recurrence risk prediction models for hepatocellular carcinoma after liver transplantation, *J. Gastroenterol. Hepatol.* 39 (2024) 2272–2280.
- [5] R. Nevola, R. Ruocco, L. Criscuolo, et al., Predictors of early and late hepatocellular carcinoma recurrence, *World J. Gastroenterol.* 29 (2023) 1243–1260.
- [6] H. Chen, D. Lu, X. Yang, et al., One shoot, two birds: alleviating inflammation caused by ischemia/Reperfusion injury to reduce the recurrence of hepatocellular carcinoma, *Front. Immunol.* 13 (2022) 879552.
- [7] J. Liu, K. Man, Mechanistic Insight and clinical implications of Ischemia/Reperfusion Injury Post Liver Transplantation, *Cell. Mol. Gastroenterol. Hepatol.* 15 (2023) 1463–1474.
- [8] S. Nagai, A. Yoshida, M. Facciuto, et al., Ischemia time impacts recurrence of hepatocellular carcinoma after liver transplantation, *Hepatology* 61 (2015) 895–904.
- [9] A. Kornberg, U. Witt, J. Kornberg, et al., Extended ischemia times promote risk of HCC recurrence in liver transplant patients, *Dig. Dis. Sci.* 60 (2015) 2832–2839.
- [10] S. Canizares, A. Montalvan, R. Chumdermpadetsuk, et al., Liver machine perfusion technology: expanding the donor pool to improve access to liver transplantation, *Am. J. Transplant.* 24 (2024) 1664–1674.

- [11] M. Mueller, M. Kalisvaart, J. O'Rourke, et al., Hypothermic oxygenated liver perfusion (HOPE) prevents tumor recurrence in liver transplantation from donation after circulatory death, *Ann. Surg.* 272 (2020) 759–765.
- [12] Y. Tang, T. Wang, W. Ju, et al., Ischemic-Free liver transplantation reduces the recurrence of hepatocellular carcinoma after liver transplantation, *Front. Oncol.* 11 (2021) 773535.
- [13] F. Rigo, N. De Stefano, D. Patrono, et al., Impact of hypothermic oxygenated machine perfusion on hepatocellular carcinoma recurrence after liver transplantation, *J. Personalized Med.* 13 (2023).
- [14] L.E. Matevish, J. Guo, A.D. Shubin, et al., Transplantation of patients with hepatocellular carcinoma through increased utilization of machine perfusion technology, *Transplant Direct* 11 (2025) e1777.
- [15] Z. Xiang, Y. Song, J. Li, et al., Impact of donor CMV-Seropostive grafts on prognosis of liver transplantation for hepatocellular carcinoma, *J. Med. Virol.* 97 (2025) e70150.
- [16] X. Huang, H. Wen, Y. Huang, et al., Preoperative circulating tumor cells indicate microvascular invasion and dynamical detection indicate the early recurrence of hepatocellular carcinoma, *BMC Cancer* 25 (2025) 1025.
- [17] L. Rauter, D. Kollmann, J. Schiefer, et al., Endothelial glycocalyx damage marker syndecan-1 during hypothermic oxygenated machine perfusion of donor grafts facilitates prediction of early allograft dysfunction after liver transplantation, *Hepatobiliary Surg. Nutr.* 14 (2025) 233–245.
- [18] R. Panconesi, M.F. Carvalho, J. Eden, et al., Mitochondrial injury during normothermic regional perfusion (NRP) and hypothermic oxygenated perfusion (HOPE) in a rodent model of DCD liver transplantation, *EBioMedicine* 98 (2023) 104861.
- [19] Q. Wang, P. Yu, C. Liu, et al., Mitochondrial fragmentation in liver cancer: emerging player and promising therapeutic opportunities, *Cancer Lett.* 549 (2022) 215912.
- [20] C. Uslu, E. Kapan, A. Lyakhovich, Cancer resistance and metastasis are maintained through oxidative phosphorylation, *Cancer Lett.* 587 (2024) 216705.
- [21] L. Masior, M. Grał, Methods of attenuating ischemia-reperfusion injury in liver transplantation for hepatocellular carcinoma, *Int. J. Mol. Sci.* 22 (2021).
- [22] S.Q. Sin, C.D. Mohan, R.M.W. Goh, et al., Hypoxia signaling in hepatocellular carcinoma: challenges and therapeutic opportunities, *Cancer Metastasis Rev.* 42 (2023) 741–764.
- [23] Z. Xiang, J. Li, Y. Xu, et al., Cell differentiation-related signaling pathways in hepatocellular carcinoma metastasis, *Cancer Lett.* 627 (2025) 217846.
- [24] M. Maspero, S. Yilmaz, B. Cazzaniga, et al., The role of ischaemia-reperfusion injury and liver regeneration in hepatic tumour recurrence, *JHEP Rep* 5 (2023) 100846.
- [25] K.B. Garcia, A. Hussein, S. Satish, et al., Machine perfusion as a strategy to decrease ischemia-reperfusion injury and lower cancer recurrence following liver transplantation, *Cancers (Basel)* 16 (2024).
- [26] C.J. Wehrle, R. Raj, M. Maspero, et al., Risk assessment in liver transplantation for hepatocellular carcinoma: long-term follow-up of a two-centre experience, *Int. J. Surg.* 110 (2024) 2818–2831.
- [27] L.A. Orci, T. Berney, P.E. Majno, et al., Donor characteristics and risk of hepatocellular carcinoma recurrence after liver transplantation, *Br. J. Surg.* 102 (2015) 1250–1257.
- [28] J. Yu, T. Yunhua, Y. Guo, et al., Beyond graft function impairment after liver transplantation: the prolonged cold ischemia time impact on recurrence of hepatocellular carcinoma after liver transplantation-a single-center retrospective study, *PeerJ* 12 (2024) e18126.
- [29] C. Xie, N. Qiu, C. Wang, et al., G-LERP/miR-374i-b Attenuates IRI and suppresses hepatocellular carcinoma progression, *Transplantation* (2025).
- [30] W. Shu, S. Xu, S. Zheng, et al., Does the utilization of machine perfusion require a boundary for livers donated after circulatory death? *Liver Transplant.* (2024).
- [31] A. Parente, F. Tirota, A. Pini, et al., Machine perfusion techniques for liver transplantation - a meta-analysis of the first seven randomized-controlled trials, *J. Hepatol.* 79 (2023) 1201–1213.
- [32] O.B. van Leeuwen, D. Nasralla, C.D.L. Ceresa, et al., The IDEAL framework for machine perfusion in liver transplantation, *Nat. Rev. Gastroenterol. Hepatol.* (2025).
- [33] K. Sun, C. Jiao, R. Panconesi, et al., Quantifying Flavin mononucleotide: an internationally validated methodological approach for enhanced decision making in organ transplantation, *EBioMedicine* 116 (2025) 105761.
- [34] A. Schlegel, H. Mergental, C. Fondevila, et al., Machine perfusion of the liver and bioengineering, *J. Hepatol.* 78 (2023) 1181–1198.